✅ 1. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions and equations, and geometric representations

- Highest relevance.
- The core of this task lies in manipulating geometric representations:
  - Students must interpret the grid diagram to distinguish multiple regions (entire shop, counter, service area, seating area, entrance).
  - They must decompose a complex shape or use a "whole minus parts" strategy to find the desired area.
  - They carefully count grid squares (possibly dealing with partial squares or irregular edges through estimation or recombination).
- Simultaneously, they manipulate numerical data: converting square counts to real area via multiplication by 0.25 m^2 per square.
- This fully demonstrates integrated competence in handling both spatial forms and quantitative data.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving the problem requires a clear multi-stage process. Two common strategies are:Strategy A (Total minus excluded areas):

  1. Count total squares in the entire shop;
  2. Count squares in the service area + counter;
  3. Subtract to get usable area;
  4. Multiply by 0.25 m^2 to get total floor area.

  Strategy B (Direct summation of included areas):

  1. Count squares in the entrance area and seating area separately;
  2. Add them together;
  3. Multiply by 0.25 m^2.
- Regardless of approach, the solution involves multiple logical steps, intermediate calculations, and unit conversion, precisely matching Competency #12.

✅ 3. Competencies #4 and #6: Make/use mathematical diagrams and extract mathematical information from them

- Highly relevant.
- Although students don't create a new diagram, they must deeply engage with the given grid as a mathematical tool:
  - Accurately distinguish functional zones (labeled "Entrance area," "Seating area," "Counter," "Service area");
  - Mentally or physically mark, group, or shade target regions;
  - Extract area-related information through square counting.
- This exemplifies the ability to transform a real-world layout into a quantifiable mathematical object, a hallmark of Competencies #4 and #6.

✅ 4. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Explicitly required.
- The prompt states: "Show your work."
- A strong response includes:
  - A brief description of the chosen strategy (e.g., "I subtracted the service area from the total shop area");
  - Key intermediate results (e.g., "Total shop: 80 squares; Service area + counter: 28 squares; Usable area: 52 squares");
  - Unit conversion ($52 \times 0.25 = 13$ m²);
  - Final answer with units ("Total flooring area needed: 13 m²").
- This transparent, structured communication is exactly what Competency #8 emphasizes.

✅ 5. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process (e.g., working to a context-appropriate level of accuracy)

- Moderately relevant.
- Contextual understanding informs critical decisions:
  - Clearly interpreting "excluding the service area and the counter"—these adjacent, counter-enclosed zones must be accurately bounded;
  - Given the grid resolution of 0.5 m, area precision to 0.25 m² is sufficient; fine estimation of diagonal cuts (e.g., half-squares) can be reasonably approximated;
  - Mistaking walls or outdoor space as part of the shop would reflect poor contextual interpretation.
- Thus, realistic knowledge of shop layout helps correctly define the region to be calculated.

✅ 6. Competency #1: Perform a simple calculation

- Basic but necessary.
- Involves integer addition/subtraction and multiplication (e.g., 52×0.25). While straightforward, errors in counting invalidate the entire solution.
- This is a supporting skill, not the primary challenge.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 本题核心是对几何表示的操作：
  - 学生需在网格图上识别多个区域（整个店铺、服务台、服务区域、座位区、入口区）；
  - 将复杂形状分解或通过“整体减去部分”来求解；
  - 精确数出所需区域的方格数量（可能涉及半格、不规则边缘的估算或重组）；
- 同时需操作数值：将方格数 × 0.25 m^2/格，得到实际面积。
- 这一过程充分体现了对空间图形与数值数据的综合操作能力。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需遵循明确的多阶段流程，常见策略有两种：
  策略 A（整体减去排除部分）：
  1. 数出整个店铺的总方格数；
  2. 数出服务区域 + 服务台的方格数；
  3. 相减得可用面积；
  4. 乘以 0.25 m^2 得总面积。

  策略 B（直接相加保留区域）：
  1. 分别数出入口区和座位区的方格数；
  2. 求和；
  3. 乘以 0.25 m^2。
- 无论哪种策略，都涉及多个逻辑步骤、中间计算和单位转换，完全符合能力 #12 的要求。

✅ 3. 能力 #4 与 #6：制作/使用数学图表并从中提取数学信息

- 高度相关。
- 虽然学生不需绘制新图，但必须深度利用给定的网格图作为数学工具：
  - 准确区分不同功能区域（图中标注了“Entrance area”、“Seating area”、“Counter”、“Service area”）；
  - 在脑中或纸上标记、分组或着色目标区域；
  - 从视觉表示中提取面积信息（通过计数方格）。
- 这体现了将现实情境图转化为可量化数学对象的能力，是能力 #4 和 #6 的典型应用。

✅ 4. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 题目明确要求。
- 题干强调：“请展示你的计算过程。”
- 优秀回答应包含：
  - 所选策略说明（如“我用总面积减去服务区域”）；
  - 关键中间结果（如“总店铺：80 格；服务区域+柜台：28 格；可用区域：52 格”）；
  - 单位换算（52 × 0.25 = 13 m²）；
  - 最终答案带单位。
- 这种透明、结构化的表达正是能力 #8 的核心。

✅ 5. 能力 #13：利用对情境的理解来指导或加速数学解题过程（如采用符合情境的精度）

- 中等相关。
- 情境理解影响关键决策：
  - 明确“不包括服务区域和服务台”——这两个区域通常相邻且被柜台围合，需准确识别边界；
  - 网格分辨率为 0.5 m，因此面积精确到 0.25 m² 即可，无需更细估算（如对角线切割的半格可合理近似）；
  - 若误将墙体或外部空间计入，即为情境理解失误。
- 因此，真实店铺布局知识有助于正确界定计算范围。

✅ 6. 能力 #1：执行简单计算

- 基础但必要。
- 涉及整数加减、乘法（如 52 × 0.25），虽简单，但若计数错误则全盘皆错。
- 属于支撑性能力，非核心难点。